Problem Set 3: Multiple Linear Regression

POLS 602

Instructions

Submit both your code and answers by pushing them to your github repo. They may be in separate files, or in a single file like a .qmd, .rmd, or .ipynb notebook. Comment your code and make sure it executes without errors.

Part 1: Paper Analysis

Some studies have found that when reading academic papers, more experienced researchers tend to focus on the methods and results sections, while undergraduates and junior researchers focus more on the paper's narrative. Evaluating research empirically is a skill that takes time to develop, and this exercise is designed to help you cultivate that skill.

Evaluate Canes-Wrone, Brady, and Cogan's (2002) APSR paper, "Out of Step, Out of Office: Electoral Accountability and House Members' Voting" and answer the following questions. I encourage you to complete this part of the exercise with classmates.

1. Research Goals. Is the goal of the study causal inference, description, prediction, or something else? Have the authors clearly stated their goals? Describe any strengths or weaknesses in how the authors articulate their research objectives.

2. Estimands. Have the authors sufficiently defined their theoretical and empirical estimands? Discuss what these estimands are and explain how the authors could clarify them if necessary.

3. Identification Strategy. The way you connect your theoretical estimand to your empirical estimand is known as identification. In other words, what does the research do to ensure that the empirical estimand is a good measure of the theoretical estimand? Describe the authors' identification strategy.

4. Assessment of Findings. Provide an overall assessment of the paper and its conclusions. Does the identification strategy support the authors' claims? For example, could the regression coefficients be credibly interpreted as causal effects if causal inference is the goal? Does the model adequately represent the real-world data-generating process? Does the data credibly measure the phenomena being studied?

5. Broader Contribution. Despite any weaknesses, can this research still inform our understanding of the world? If so, how?

Part 2: Data Analysis

In this part, you will use fixed effects to control for unobserved differences between groups, and examine both the benefits and the limitations of doing so.

A common way to measure how people feel about other groups is the feeling thermometer. Respondents rate how "warm" they feel toward a group on a scale from 0 to 100, where 100 is very warm and 0 is very cold. The file `thermometers_hw03.csv` contains feeling thermometers for several groups, along with demographic information about the respondents. The survey was taken in 2017. The file also contains `neighborhood`, a variable that was simulated for this assignment and is not part of the original survey.

Our dependent variable will be the police thermometer (`ft_police`), our independent variable will be the Muslim thermometer (`ft_muslim`), and the grouping variable is `party_id` unless a question says otherwise. The pooled model is a regression that ignores group membership.

(a) Load the data. Keep only respondents with non-missing values of both `ft_police` and `ft_muslim`, so that every model below is estimated on the same sample. Convert `party_id` to a factor with Democrat as the reference category.

(b) Regress `ft_police` on `party_id`. Interpret the intercept and each coefficient. Show how to recover the sample mean of `ft_police` in each party from the estimated coefficients.

(c) Estimate two models: the pooled regression of `ft_police` on `ft_muslim`, and the regression of `ft_police` on `ft_muslim` and `party_id` (the party fixed effects model). Report the coefficient on `ft_muslim` in each model. Explain why the coefficients differ.

(d) For each respondent, subtract the mean of `ft_police` in the respondent's party from the observed value of `ft_police`. Do the same for `ft_muslim`. Regress the demeaned `ft_police` on the demeaned `ft_muslim`. Compare the slope to the coefficient on `ft_muslim` in the party fixed effects model from (c). What share of the total variance of `ft_muslim` is within-party variance? Does the fixed effects estimator use the total variance or the within-party variance of `ft_muslim` to estimate the coefficient? What happens to the between-party variance?

(e) Estimate the pooled model, and then a model that adds `neighborhood` as fixed effects. Compare the coefficient on `ft_muslim` and its standard error across the pooled model, the party fixed effects model, and the neighborhood fixed effects model. Explain why the standard error differs across the three models, and what this shows about the cost of including fixed effects. Hint: consider your answer about the total variance and the within-party variance of `ft_muslim` in (d).

(f) Return to the party fixed effects model and add `educ`, `race`, and `sex` as control variables. Report how the coefficient on `ft_muslim` changes. Explain what a change in the coefficient implies about whether the party fixed effects controlled for confounding by these variables. State what you could conclude, and what you could not conclude, if the coefficient had not changed.

(g) The party fixed effects model allows the intercept to differ by party but constrains the slope on `ft_muslim` to be the same in every party. Estimate a model that allows the slope to differ by party (an interaction between `ft_muslim` and `party_id`), and report the estimated slope for each party. Test the null hypothesis that the slope is the same in every party using an F-test. Compare the pooled, party fixed effects, and interaction models using R-squared and RMSE. Based on these results, is the common-slope restriction supported? Distinguish between statistical significance and the size of the differences in your answer.